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Grain yield, symbiotic N_2 fixation and interspecific competition for inorganic N in pea-barley intercrops

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Abstract

The effect of mixed intercropping of field pea (*Pisum sativum* L.) and spring barley (*Hordeum vulgare* L.), compared to monocrop cultivation, on the yield and crop-N dynamics was studied in a 4-yr field experiment using ^{15}N-isotope dilution technique. Crops were grown with or without the supply of 5 g ^{15}N-labeled N m^{-2}. The effect of intercropping on the dry matter and N yields, competition for inorganic N among the intercrop components, symbiotic fixation in pea and N transfer from pea to barley were determined. As an average of four years the grain yields were similar in monocropped pea, monocropped and fertilized barley and the intercrop without N fertilizer supply. Nitrogen fertilization did not influence the intercrop yield, but decreased the proportion of pea in the yield. Relative yield totals (RYT) showed that the environmental sources for plant growth were used from 12 to 31% more efficiently by the intercrop than by the monocrops, and N fertilization decreased RYT-values. Intercrop yields were less stable than monocrop barley yields, but more stable than the yield of monocropped pea. Barley competed strongly for soil and fertilizer N in the intercrop, and was up to 30 times more competitive than pea for inorganic N. Consequently, barley obtained a more than proportionate share of the inorganic N in the intercrop. At maturity the total recovery of fertilizer N was not significantly different between crops, averaging 65% of the supplied N. The fertilizer N recovered in pea constituted only 9% of total fertilizer-N recovery in the intercrop. The amount of symbiotic N_2 fixation in the intercrop was less than expected from its composition and the fixation in monocrop. This indicates that the competition from barley had a negative effect on the fixation, perhaps via shading. At maturity, the average amount of N_2 fixation was 17.7 g N m^{-2} in the monocrop and 5.1 g N m^{-2} in the intercropped pea. A higher proportion of total N in pea was derived from N_2 fixation in the intercrop than in the monocrop, on average 82% and 62%, respectively. The ^{15}N enrichment of intercropped barley tended to be slightly lower than of monocropped barley, although not significantly. Consequently, there was no evidence for pea N being transferred to barley. The intercropping advantage in the pea-barley intercrop is mainly due to the complimentary use of soil inorganic and atmospheric N sources by the intercrop components, resulting in reduced competition for inorganic N, rather than a facilitative effect, in which symbiotically fixed N_2 is made available to barley.

Abbreviations: MC - monocrop, IC - intercrop, PMC - pea monocrop, BMC - barley monocrop, PIC - pea in intercrop, BIC - barley in intercrop.

Introduction

Intercropping is defined as the simultaneous growing of two or more crop species on the same area of ground (Willey, 1979a). Intercropping may have several advantages, many of which have been listed by Trenbath (1976) and Willey (1979a, 1979b). Important ones are: 1) improved utilization of environmental sources for plant growth, such as light, water and nutri-

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ents, 2) higher yield of the intercrop compared to the monocrops, 3) improved yield stability over seasons compared to monocropping, and 4) reduced yield losses due to pathogens, insects and weeds.

Advantages of intercropping compared to monocrops are due to interactions between intercrop components, caused by differences in the competitive ability for environmental sources for plant growth (De Wit, 1960; Trenbath, 1976). If the intercrop components differ in their use of environmental sources in a way that they can "complement" each other in time and space, they may use environmental sources more efficiently, than when grown in monocrops and perhaps produce a higher yield than the higher yielding monocrop (Willey, 1979a). In terms of competition, this means that the components are not competing for exactly the same ecological niches and that intercrop competition is weaker than intercrop competition for a given factor (Vandermeer, 1989; Willey, 1979a).

Legumes and nonlegumes can "complement" each other in the use of N sources since both the legume and nonlegume utilize soil inorganic N sources, but the nodulated legume can also fix atmospheric N_2 in symbiosis with *Rhizobium*. Thus, the legumes may be able to cover their N requirements by N_2 fixation, if the nonlegume is a stronger competitor soil inorganic N, provided competition for other environmental factors for growth e.g. light is not limiting the fixation process (Wahua and Miller, 1978). High concentrations of soil inorganic N can inhibite N_2 fixation in legumes (Sprent and Minchin, 1985). The legume can reduce the soil inorganic N concentration in the intercrop, and in this way N_2 fixation may increase compared to the monocrop situation (e.g. Danso et al., 1987).

The most exciting aspect of N interaction in intercrops of legumes and nonlegumes is whether fixed N_2 can contribute to the N nutrition of the nonlegume. This process is often termed N transfer and in terms of competition it is considered a facilitative effect (Vandermeer, 1989). However, the quantification of N transfer between plants is not simple. The so-called indirect ^{15}N-isotope dilution method have been employed in many studies (e.g. Danso et al., 1987; Eaglesham et al., 1981). Using this method, the soil inorganic N pool is labeled with ^{15}N and by comparing the ^{15}N enrichments of the nonlegume in the intercrop and monocrop situations, an estimate of the N transferred to the nonlegume can be obtained. Evidence for N-transfer in annual grain legume-nonlegume intercrops have been observed only in a few studies (e.g. Eaglesham et al., 1981), whereas numerous studies have not shown evidence of significant N transfer, e.g. Cowell et al. (1989).

Field pea (*Pisum sativum* L.) can be intercropped with cereals and other nonlegumes in temperate cropping systems. The productivity of such intercrops has been investigated (e.g. Cowell et al., 1989; Izaurralde et al, 1992; Johnston et al., 1978; Waterer et al., 1994). However, only few studies have focussed on the N dynamics in temperate grain legume-cereal intercrops.

The aims of this work were to determine: 1) the productivity and stability of pea-barley intercrop yields as compared to monocrops in four cropping season as influenced by N fertilization, 2) the competition for soil and fertilizer N among intercrop components and the crop recovery of fertilizer N, 3) the effect of intercropping on N_2 fixation in pea, and 4) the relative effects of reduced competition for soil inorganic N sources and N transfer in the improved N nutrition of intercropped barley plants.

Materials and methods

Site and soil

The experiments were carried out in 1980-82 and 1984 in the experimental fields at Risø National Laboratory, Denmark (55° 41′ N, 12° 05′ E). The amounts of rainfall in each growth season are shown in Table 1. The 25-yr mean annual rainfall at Risø is 550 mm.

The soil was a sandy loam (Typic Hapludalf) with 11% clay, 14% silt, 49% fine sand and 25% coarse sand, a pH of 7.0 in water, 1% total C and 0.1% total N in the topsoil (0-20 cm). White mustard (*Sinapis alba* L.) was grown in the preceding year to each experiment. The soil contained efficient populations of native *Rhizobium leguminosarum* bv. *viciae*. Two days before sowing, the soil was sampled from the 0-20 and 20-40 cm soil layers and analysed for KCl-extractable inorganic N.

Experimental design

Field pea (*Pisum sativum* L., cv. Bodil), which is an early, white-flowered, dwarf and determinate cultivar and spring barley (*Hordeum vulgare* L. cv. Nery) a medium early, short cultivar, were grown as monocrops (PMC, BMC) and in a mixed intercrop (IC). Crops were in grown with (+N) or without (−N) the supply of 5 g N m^{-2}. The fertilizer N was labeled with ^{15}N to determine fertilizer N uptake, soil N uptake, N_2

Table 1. Accumulated monthly rainfall (mm) at Risø during the growth seasons and the 25-yr average

Year	Apr	May	Jun	Jul	Aug	∑ Season
1980	19	6	57	93	50	225
1981	18	59	58	102	47	284
1982	10	66	105	17	54	252
1984	11	27	107	22	44	211
25-yr mean	27	37	39	57	57	217

fixation and N transfer from pea to barley. The experimental layouts were two (1980-82) or nine (1984) randomized split-plot designs with crops as main plots, N fertilization as subplots, and three replicates in 1980-82 and four replicates in 1984. Each subplot (5.4 m^2) consisted of ten rows of lenght 4.4 m spaced 15 cm apart.

Crop management practices

The experimental areas were fertilized by broadcasting 30 kg P ha^{-1} and 50 kg K ha^{-1} prior to sowing. Fungicide treated (Thiram 80) seeds were sown on 16 April 1980, 13 April 1981, 7 April 1982 and 16 April 1984 using a ten-rowed drill. The target monocrop plant populations were 350 barley plants and 80 pea plants per m^2. The intercrop composition was based on the replacement principle (De Wit and van den Bergh, 1965). The rationale for using this principle is that the total plant density of the intercrop equals the total density of monocrops (Trenbath, 1976). Thus, it is ensured that the interactions between intercrop components is not confounded by alterations in the plant density in the intercrop compared to monocrops (De Wit, 1960). The plant population of each species in the monocrop is assigned a density of 1.0; each component in the intercrop is then planted at a given fraction of its monocrop population and the sum of fractions in the intercrop equals 1.0. The target intercrop density was 0.5 of the monocrop density of each species. Barley and pea seeds were mixed before sowing. The number of seedlings in four rows of lenght 1 m was counted two weeks after seedling emergence (Table 2).

A ^{15}N microplot with 6 rows of 1.2 m (1.08 m^2) was placed within each subplot receiving N fertilizer. The ^{15}N-labeled N was a mixture of $Ca(NO_3)_2$ and KNO_3 with ^{15}N enrichments of 1.0 to 3.9 atom % ^{15}N excess. The ^{15}N enrichment of the fertilizer N varied according to lenght of the growth period, the longer the growth before growth analysis, the higher the enrichment. One week after seedling emergence, the labeled N was added as an aqueous solution by spraying in a grid on the microplot. Unlabeled $Ca(NO_3)_2$ was applied to the remainder of the plot. Weeds and leaf eating weavills (*Sitona lineatus*) were controlled with appropriate pesticides.

Plant sampling and analytical methods

Above-ground plant parts were harvested by cutting the plant 2 cm above the soil surface. In 1980-1982 two harvests were made each growth season; the first harvest was taken at the full bloom/flat pod growth stage in pea, corresponding to one week after anthesis (heading) in barley (7-8 weeks after seedling emergence). The second harvest was taken at maturity (13-15 weeks after seedling emergence). In 1984 crops were harvested nine times: 3, 5, 6, 8, 9, 10, 11, 13, 15 weeks after seedling emergence.

The harvested plot area was 2.2 m^2. Two central rows of 40 cm within in the ^{15}N microplots were harvested before the remainder part of the plot was harvested. Intercrops were separated in barley and pea. At the first harvest in 1980-82 and the first four harvests 1984, above-ground plant parts were not sub-divided. At harvest five to eight in 1984, above-ground plant parts were subdivided between pods and vegetative parts for pea and between heads and vegetative parts for barley. At the final harvest, pods were subdivided in grains and pod walls, and ears into grains and glume. Each component was weighed and subsamples taken for determination of dry matter content after drying at 80 °C for 24 h.

Total N was determined on ground samples by a semi-micro Kjeldahl method which included nitrate. ^{15}N analysis was carried out using the modified Dumas sample preparation and isotope ratio mass spectrometry (Fiedler and Proksch, 1975). Ammonium

Table 2. Plant populations in mono- and intercrops of pea and barley two weeks after seedling emergence

Year	Plant population[a] (plants m^{-2})				Intercrop composition[b]	
	PMC	BMC	PIC	BIC	% Pea	% Barley
1980	63±1	275±3	30±1	154±1	46	54
1981	65±1	381±9	35±2	185±2	53	47
1982	88±1	299±2	45±3	167±3	48	52
1984	88±2	356±2	48±1	178±2	49	51

[a]PMC-Pea monocrop, BMC-Barley monocrop, PIC-Pea in intercrop, BIC-Barley in intercrop.
[b]Calculation based on plant units, e.g. in 1980 1 unit = 1 pea plant or 275/63=4.2 barley plants.

for $^{15}N/^{14}N$ analysis was obtained by distillation of the Kjeldahl digest and collection in 0.5 *N* H_2SO_4. Aliquots corresponding to approximately 1 mg were evaporated to dryness in ampoules of Jena glass and combusted with CaO and CuO at 560 °C for 3 hrs in the evacuated ampoule. The $^{15}N/^{14}N$ ratio of the N_2 in ampoules was measured at a MAT CH 4/5 mass spectrometer equipped with a thermostated gas-inlet system with molecular flow into the ion source (Egsgaard et al., 1989)

Nitrate and ammonium were extracted from 10 g of fresh soil with 100 mL 2 *M* KCl and determined by distillation (Keeney and Nelson, 1982).

Calculations and statistics

The atom % ^{15}N excess was obtained by substracting the natural ^{15}N abundance in unfertilized barley (0.370 atom % ^{15}N) from the ^{15}N abundance in the other treatments. When above-ground plant parts were separated before ^{15}N analysis, a weighted atom % ^{15}N excess for the total above-ground N was calculated. The ^{15}N enrichments were corrected for seed N assuming that 50% of the barley (1 mg N $seed^{-1}$) and pea (10 mg N $seed^{-1}$) seed N were present in above-ground plants parts.

Nitrogen in crops derived from ^{15}N-labeled fertilizer N (Ndff), soil N (Ndfs), and fixation in pea (Ndfa) was calculated using well-known ^{15}N-isotope dilution equations (Fried and Middelboe, 1977). Monocropped barley was used as reference crop for calculating N_2 fixation in mono- and intercropped pea.

The relative yield total (RYT_{BP}) for intercrops of barley and pea was calculated according to De Wit and van den Bergh (1965). The RYT_{BP} for an intercrop of barley and pea is the sum of the RY-values for barley and pea:

$$RY_B = Y_{BP}/Y_{BB} \quad (1)$$

$$RY_P = Y_{PB}/Y_{PP} \quad (2)$$

$$RYT_{BP} = RY_B + RY_P \quad (3)$$

where Y_{BP} and Y_{PB} are yields (dry matter, total N, fertilizer N, etc.) of barley and pea, respectively, in the intercrop and Y_{BB} and Y_{PP} are the yields of barley and pea in monocrops, respectively. A RYT > 1 indicate an advantage from intercropping in terms of environmental resource use. However, the intercrop may not necessarily be higher yielding than the more productive of the monocrops, but RYT > 1 is a prerequiste for this. If RYT < 1, then environmental sources for plant growth are used more efficiently by monocrops than by the intercrop. Calculation of the Land Equivalence Ratio (LER) will give the same values as RYT, when using the replacement design (Trenbath, 1976).

The competitive ability of barley for fertilizer and soil N realtive to pea was evaluated by calculating the competitive ratio (CR_{BP}) of barley with respect to pea (Willey and Rao, 1980). This ratio indicates the number of times by which barley is more competitive than pea and is defined as:

$$CR_{BP} = \frac{RY_B}{RY_P} \times \frac{\%\text{ pea in intercrop}}{\%\text{ barley in intercrop}} \quad (4)$$

Nitrogen balances for crops were determined to evaluate the effect of cropping on soil N. Since the root N was not determined, balances represent minimum values for the PMC and IC, because fixed N was also present in roots and rhizodeposition. Balances were only calculated for crops supplied with labeled N.

$$\text{N balance} = \text{Fertilizer N} + N_2 \text{ fixation - grain N} \quad (5)$$

Analysis of variance was carried out on data using the ANOVA and GLM procedures in SAS (SAS, 1990).

Table 3. Grain yields of pea and barley grown in mono- and intercrops, % pea DM in the intercrop (IC) and the combined monocrop (Comb. MC) grain yields and the Relative Yield Total (RYT)

Year	N fertilizer treatment	Crop			% pea in		RYT
		PMC	BMC	IC	IC[b]	Comb.MC[c]	
		(g DM m^{-2})			(%)	(%)	
1980	−N	456	412	479	29	53	1.13
	+N	357	550	541	15	39	1.06
		LSD(0.05) = 65			*		
1981	−N	437	355	509	46	55	1.31
	+N	456	467	519	15	49	1.13
		LSD(0.05) =104			***		
1982	−N	714	473	658	47	60	1.17
	+N	767	615	665	30	55	1.02
		LSD(0.05) = 63			NS		
1984	−N	659	332	561	67	66	1.12
	+N	661	579	607	34	53	1.00
		LSD(0.05) = 61			***		
Mean	−N	567	393	552	47	59	1.18
	+N	560	553	583	24	50	1.05
CV[a]	−N	24	16	14	33	ND	7
	+N	3	11	11	42	ND	5

[a]CV: Coefficient of variation (%) on average grain yields.
[b]Differences between means within each year are significant at the respective probability levels. * : $p \leq 0.05$, ***: $p \leq 0.001$. NS: Not significant. ND: Not determined.
[c]% pea in combined monocrop yield = PMC yield × 100/(PMC yield + BMC yield).

Analysis of variance was not carried out on soil N uptake in the crops, because the estimates of soil N uptake in pea is not independent of the soil N uptake in the barley monocrop.

Results

Grain yield, resource use and yield stability

The intercrop was the higher yielding crop in 1980 and 1981, whereas the pea monocrop had the highest yields in 1982 and 1984 (Table 3). As an average of four years, intercrops without fertilization produced a similar grain yield as the pea and the barley crop with N fertilizer supply (mean: 558 g m^{-2}). Nitrogen fertilization increased the grain yield of monocrop barley, but had no effect on the grain yield of pea grown in monocrop (Table 3). Fertilizer N increased the barley and reduced the pea yield in the intercrop. As a result, the total intercrop grain yield was not significantly influenced by fertilizer N, but the proportion of pea in the intercrop yield was appproximately halved (Table 3). The calculation of the per cent of pea in the combined monocrop yield demonstrated that N fertilization reduced the per cent of pea the intercrop. If there was no competitive interaction between pea and barley, it would be expected that the % pea in the intercrop yield equaled the % pea in the combined monocrop yield (Table 3).

Thus, barley was the dominant component in the intercrop. Without N fertilizer the relative yield totals (RYT) varied between 1.12 and 1.31, indicating a much better utilization the environmental sources for plant growth by the intercrop than by the monocrops (Table 3). Nitrogen fertilization increased the compet-

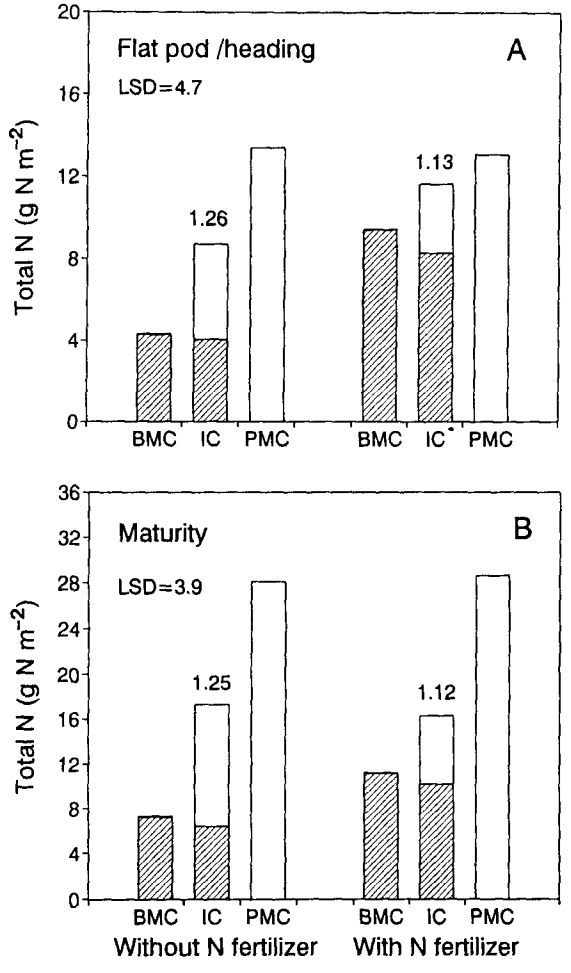


Figure 1. Accumulation of total N in crops at (**A**) the flat pod/heading growth stage and (**B**) at maturity. Hatched part of column represent barley N; empty part represent pea N. Values on top of IC columns are RYT_N.

itive ability of barley in the intercrop and decreased the RYT values, because the RY_P's were reduced more than and the RY_B's increased (data not shown).

Grain yields were significantly ($p \leqslant 0.001$) lower in 1980 and 1981 than in 1982 and 1984 (Table 3). The yield stability over the four experimental years was evaluated by calculating the coefficient of variation (% CV) on the mean yields. The stability of the barley monocrop and the intercrop was similar, and larger than of the pea grown in monocrop.

N accumulation and grain-N yield

The pea monocrop accumulated significantly more N than the barley monocrop and the intercrop (Fig. 1). Barley accumulated similar amounts of N in the mono- and intercrop, even though the barley plant density in the intercrop was only 50% of the density in monocropped barley (Fig. 1). Significant ($p \leq 0.01$) crop × N-fertilizer interaction was found, because N fertilization increased the N accumulation in barley, but had no effect on total N accumulated in the pea and the intercrop.

Table 4. Grain N yield of pea and barley grown in mono- and intercrops and the % pea N in the intercrop grain N yield

Year	N-treatment	Crop			% pea N in intercrop N yield[a]
		PMC	BMC	IC	
		($g\ N\ m^{-2}$)			
1980	−N	18.0	6.5	11.3	47
	+N	14.5	9.5	11.6	26
		LSD(0.05) = 2.1			**
1981	−N	16.8	5.3	12.9	64
	+N	15.6	7.5	9.9	27
		LSD(0.05) = 3.1			
1982	−N	27.2	6.9	17.0	68
	+N	28.6	9.2	15.4	50
		LSD(0.05) = 2.2			*
1984	−N	24.6	4.7	17.1	81
	+N	25.1	7.8	14.2	52
		LSD(0.05) = 2.4			***
Mean	−N	21.7	5.9	14.6	65
	+N	21.0	8.5	12.8	39

[a]Differences between means within each year significant at the respective probability levels. $p \leq 0.05$, **: $p \leq 0.01$, ***: $p \leq 0.001$.

Nitrogen accumulation in monocropped pea more than doubled from the flat pod growth stage to maturity. Similarly, the N accumulated in the intercrop grown without N fertilization doubled, whereas it only increased with 40% in the intercrop supplied with fertilizer N. N fertilizer reduced the proportion of pea N in the total intercrop N yield (Fig. 1).

The RYT_N values indicated that N was used from 12 to 26% more efficiently in the intercrop than in the monocrops. The N fertilization decreased the RYT_N values from 1.25 to 1.12 at maturity (Fig. 1).

The average N concentration in pea and barley grain was 3.75 and 1.61 %N, respectively (data not shown). Intercropping significantly increased the N concentration in barley grain (mean increase: 13%). The increase

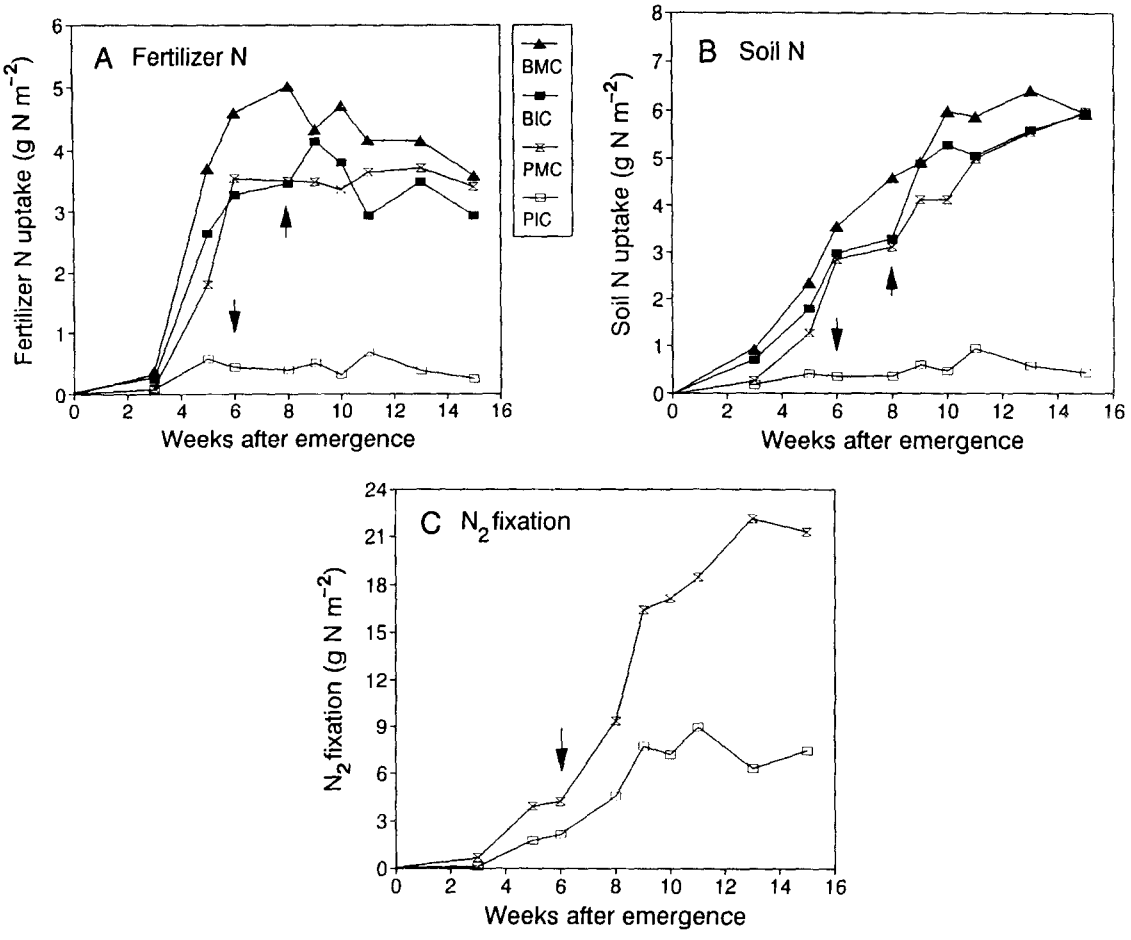


Figure 2. Cumulative fertilizer-N uptake (**A**), soil-N uptake (**B**) and symbiotic N_2 fixation (**C**) in monocrops and intercrop of pea and barley in 1984. ↓: start of flowering in pea, ↑: heading in barley. Values are mean of 4 replicates.

in barley grain N concentration due to intercropping, was larger than the increase due to N fertilization (data not shown).

The grain-N yield of the intercrop was intermediate to the monocrop yields (Table 4). Due to a higher N concentration in pea grain the proportion of pea N in the total grain-N yield was higher than the proportion of pea in the dry matter yield (Table 3 and 4). As an average of 4 years, N fertilization decreased the intercrop grain-N yield by 14%.

Uptake of fertilizer N

The sequential harvesting in 1984 showed that the major part of the fertilizer N was taken up from 3 to 6 weeks after seedling emergence (Fig. 2A). The amount of fertilizer N accumulated in the pea monocrop was maintained at a constant level (3.4-3.7 g labeled N m^{-2}) during reproductive growth, whereas it declined from 5.0 to 3.6 g labeled N m^{-2} in the the barley crop (Fig. 2A).

Intercropped barley competed strongly for the labeled fertilizer N, especially after the second harvest (Fig. 2A). Maximum recovery of fertilizer N in the intercrop was found to 93±6.5% (±SD) at the fifth harvest, but due to a subsequent decline in the recovery in both pea and barley, only 63% (mean of 4 years) was recovered at maturity (Fig. 3).

A comparison of the recovery of fertilizer N at the flat pod/heading growth stage in the three crops showed that the total recovery of fertilizer N was similar in the barley monocrop and the intercrop and higher than in the pea monocrop (Fig. 3). The amount of fertilizer N recovered by pea in the intercrop constituted only 8.7% of the total amount of fertilizer N recovered. At maturity the recovery of fertilizer N in the three crops was not significantly different (Fig. 3).

Table 5. Relative Yields for fertilizer N, soil N and N_2 fixation in pea, RYT and the competitive ratio of barley relative to pea (C_{BP}) for fertilizer and soil N in the intercrop

Year	Fertilizer N			Soil N			N_2 fixation
	RY_P	RYT	C_{BP}	RY_P	RYT	C_{BP}	RY_P
Flat pod/heading							
1980	0.08	1.06	10.8	0.07	0.94	10.4	0.39
1981	0.05	1.02	21.0	0.06	1.04	19.1	0.26
1982	0.14	0.94	5.4	0.11	1.06	6.8	0.43
1984	0.11	0.80	6.0	0.11	0.83	6.2	0.49
Maturity							
1980	0.06	0.96	13.4	0.05	1.03	15.7	0.24
1981	0.04	1.04	29.7	0.06	1.06	18.2	0.18
1982	0.09	0.93	8.5	0.14	1.05	5.9	0.33
1984	0.07	0.89	11.3	0.07	1.07	13.9	0.35

Table 6. Percent of total N derived from N_2 fixation (% Ndfa) and amount of N_2 fixation (g N m^{-2}) by pea grown in monocrop (PMC) and intercrop (IC) at the flat pod growth stage and at maturity

Year	Crop	Flat pod growth stage		Maturity	
		(% Ndfa)	(g N m^{-2})	(% Ndfa)	(g N m^{-2})
1980	PMC	31 a[a]	3.3 a	53 a	12.8 a
	IC	69 b	1.3 b	82 b	3.1 b
1981	PMC	45 a	4.6 a	62 a	15.1 a
	IC	69 a	1.2 b	79 b	2.7 b
1982	PMC	48 a	7.3 a	64 a	21.5 a
	IC	74 b	3.2 b	81 b	7.1 b
1984	PMC	57 a	9.4 a	68 a	21.3 a
	IC	82 b	4.5 b	84 b	7.4 b
Mean	PMC	45	6.2	62	17.7
	IC	73	2.6	82	5.1

[a]Means within each year followed by the same letter are not significantly different ($p \leq 0.05$).

Calculations of RY, RYT and the competitive ratio for fertilizer N in the intercrop give an indication of the sharing and utilization of fertilizer N in the intercrop compared to the monocrops. The average RYT value for the fertilizer N uptake varied between 0.80 and 1.06, with an average of 0.96 at both harvest (Table 5). This shows that the intercrop utilized the fertilizer N with the same efficiency as the monocrops. Barley was on average 11 and 16 time more competitive than pea for fertilizer N at the flat pod/heading growth stage and maturity, respectively, as revealed from calculation of the RY values and the competitive ratio (Table 5).

Uptake of soil N

The soil contained 10.1, 5.4, 9.8 and 5.3 g KCl-extractable inorganic N m^{-2} in the 0-40 cm depth in 1980, 1981, 1982 and 1984, respectively, when sampled immediately before sowing (data not shown). In agreement with this, the total uptake of soil N in above-ground plant parts of the barley monocrop in

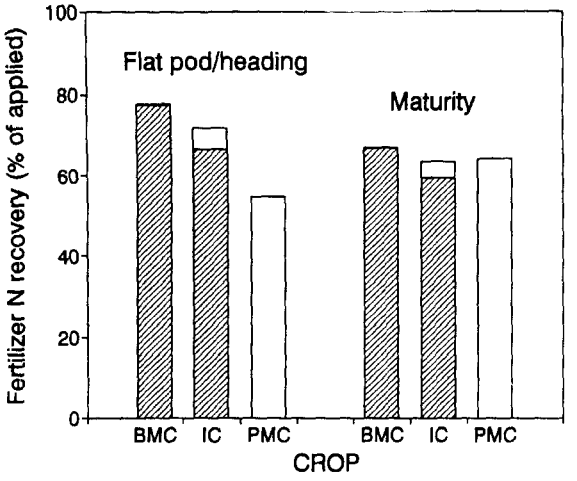


Figure 3. Recovery of fertilizer N (5 g N m^{-2}) at the flat pod/heading growth stage and at maturity in mono- and intercrops of pea and barley as mean of four experiments. Hatched part of columns represent fertilizer N recovered in barley; empty columns represent fertilizer N recovered in pea.

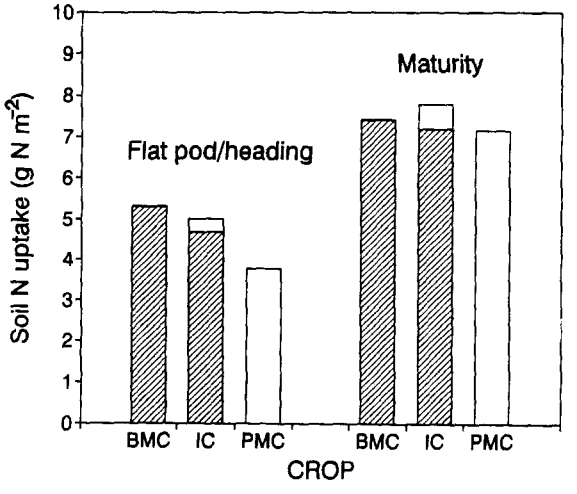


Figure 4. Soil N uptake at the flat pod/heading growth stage and at maturity in barley and pea grown in mono- and intercrops and fertilized with 5 g N m^{-2}. Hatched part of columns represent soil N in barley; empty part represent soil N in pea.

was higher in 1980 and 1982 than in 1981 and 1984 (data not shown).

The accumulation of soil N in the barley and pea monocrops and in barley grown in intercrop increased steadily during the growth season (Fig. 2B). During the vegetative growth the soil N uptake in monocropped pea was lower than in the barley monocrop and the intercrop as an average of 4 years, but at maturity no difference was found between crops (Fig. 4).

The estimated uptake of soil N in intercropped pea plants was low, due to the competition from barley

Table 7. ^{15}N enrichment of barley grown in mono- and intercrop

Harvest (weeks)	Atom % ^{15}N excess in barley	
	Monocrop	Intercrop
1980		
7	1.57	1.67
13	1.23	1.14
1981		
7	1.33	1.37
14	1.07	1.05
1982		
8	1.45	1.34
14	1.03	0.98
1984		
3	0.25	0.23
5	0.58	0.57
6	0.54	0.50
8	1.04	1.00
9	0.91	0.81
10	0.42	0.40
11	0.40	0.35
13	0.76	0.74
15	0.83	0.64

(Figs. 2B and 4). At maturity the estimated soil N uptake in intercropped pea constituted on average 7.5% of the total soil N uptake in the intercrop. At maturity total uptake of soil N in the barley and pea monocrops was very similar as an average of four years (Fig. 4). The soil N in intercropped barley also included any non-labeled N, which may have been transferred from pea to barley.

Calculation of RYT values and the competitive ratio of barley relative to pea in intercrop showed very similar patterns to those for fertilizer N (Table 5). At maturity the RYT values for soil N uptake were all higher than 1, indicating a better utilization of soil N sources (non-labeled N) by the intercrop than by the monocrops from the flat pod/heading growth stage to maturity. The competitive ratio for soil N showed that barley was up to 19 times more competitive than pea for this resource in the intercrop (Table 5).

Symbiotic N_2 fixation

The total amount of N_2 fixed symbiotically in the pea monocrop ranked 1984 $\geq$ 1982 > 1981 > 1980 (Table 6). In 1984 symbiotic N_2 fixation in intercropped pea plants was found be only slightly affected by competition from barley until 11 weeks after seedling emergence, since the relative N_2 fixation in the intercrop compared to in the monocrop (RY_P for N_2 fixation) was similar to the fraction of pea in the intercrop (Fig. 2C, Table 5). In 1980, 1981 and 1982 the amounts of N_2 fixed at the flat pod growth stage and at maturity were lower than expected from the intercrop composition (Tables 2, 5 and 6). The reduction of the RY_P for N_2 fixation from the flat pod growth stage to maturity showed that the N_2 fixation in the intercrop was further reduced during this period compared to the monocrop (Table 5). At maturity the average amount of fixed N_2 in intercropped pea was 5.1 g N m^{-2}, whereas 17.7 g N m^{-2} was fixed in the pure stand (Table 6).

The N_2 fixation process in pea was less influenced by competition from barley than the uptake of inorganic N sources, since the proportion of total N derived from fixation (%Ndfa) in the intercrop was significantly increased compared to the % Ndfa in monocropped pea. At maturity, on average 82% of the N in intercropped pea was derived from fixation, whereas fixation contributed with 62% of total N the monocrop (Table 6).

N transfer from pea to barley

At 13 out of 15 samplings, the ^{15}N enrichment of intercropped barley tended to be lower than in monocropped barley indicating that symbiotically fixed N was transferred from pea to barley (Table 7). However, due to a large variation in ^{15}N enrichments between replicate plots differences were not statistically significant at the individual harvests. Combined ANOVA's all samplings in the individual years showed that the differences between crops were not significant in 1980, 1981 and 1982. In 1984 the ^{15}N enrichments of intercropped barley combined from all harvests were significantly ($p \leq 0.05$) lower than in the monocrop. However, it is apparent that the relative differences in inter- and monocrop barley ^{15}N enrichments varied strongly with time in 1984 (Table 7) and consequently N transfer was not estimated.

Discussion

Yield advantage and stability by intercropping

The mixed intercrop of pea and barley without N fertilization utilizing environmental sources plant growth was more efficiently than monocrops. Since the RYT for the grain yield were 1.12 to 1.31 from 12 to 31% more land would have to be used in monocropping in order to obtain the same yield, if each monocrop was allocated to 50% of land. RYT values higher than unity have been reported for many intercrops of grain legumes and cereals grown under arid, semi-arid or tropical climates, see Ofori and Stern (1987). The intercrop advantage is presumed mainly to be associated with the complimentary use of N sources by components (Martin and Snaydon, 1982; Trenbath, 1976). Snaydon and Harris (1991) suggested that factors which reduce the N_2 fixation in the intercrop, such as increased concentrations of soil inorganic N and a decreased proportion of legume in the intercrop, will decrease the intercropping advantage. Thus, a negative correlation could be expected between the concentration of soil inorganic N and the RYT. However, in the present study RYT values did not correlate well with the initial concentration of soil inorganic N, indicating that other factors may also be important in determining the intercropping advantage.

Nitrogen fertilization reduced the RYT in all years, which is in agreement with other reports (e.g. Jensen et al., 1985; Martin and Snaydon, 1982; Tobita et al., 1994; Waterer et al., 1994). Supplying the intercrop with N fertilizer may reduce competition for soil inorganic N (Vandermeer, 1989), but the increased concentration of soil inorganic N, the subsequent improved growth of barley and probably increased shading of pea, may reduce N_2 fixation in intercrop and the RYT (Wahua and Miller, 1978). The decreased intercropping advantage due to N fertilization, was further underlined in the present study, where N fertilization decreased the grain-N yield of the intercrop.

The grain yield of pea was much higher in 1982 and 1984 than in 1980 and 1981. Yield differences were presumably related to the differences in the amounts of rainfall in June. Irrigation experiments have shown, that the most critical period of water stress in pea is around flowering (June) and early pod-filling (Jensen, 1987b). The evaluation of yield stability by calculating the coefficient of variation showed that the monocropped pea had the most variable yield. The monocropped barley and the intercrop yield were

much more stable than the pea crop yield. It has been suggested that an intercrop may be more stable than monocrops, since if one of the components fails, e.g. due to disease or pests, the other component may be able to use the environmental sources for plant growth, not utilized by the component which fail (Willey, 1979a).

Competition for and utilization of soil inorganic N sources

Barley was found to be much more competitive than pea for soil inorganic N sources. The competition for soil N, which was mainly for nitrate in this experiment, depend on many factors, e.g. root growth rate, root density, root lenght/root weight, time and rate of uptake (Trenbath, 1976). For a mobile nutrient-ion such as nitrate, it is expected that depletion zones around roots develop quickly, causing competition to occur much more quickly between species in an intercrop than for a non-mobile nutrient, such as phosphate (Trenbath, 1976). Due to a stronger competitive ability for soil N of grasses and cereals than of legumes (Haynes, 1980), cereals may benefit from having access to a more than proportionate share of the soil N sources in an intercrop.

The density of barley roots are much higher than of pea. Jakobsen and Nielsen (1983) found that the root density of barley (average of 8 cultivars) was on average 6 cm cm^{-3} at heading in the 0-20 cm soil layer. Dwarf and determinate pea cultivars have been reported to have root densities of 1.4 to 3 cm cm^{-3} in the top soil (Jensen, 1985, 1987a). Furthermore, the uptake of soil N take place earlier in monocropped barley than in monocropped pea (Fig. 2A). Thus, the higher density of barley than of pea roots and the earlier uptake is probably the main reasons for the much higher competitive ability of barley for soil inorganic N sources.

The higher competitive ability of barley resulted in the recovery of fertilizer N in the pea to be less than 10% of the fertilizer recovery. Tobita et al. (1994) similarly observed that pigeon peas (*Cajanus cajan* L. Millsp.) recovered only 11% of the fertilizer N recovered in pigeon pea-sorghum intercrops. The average recovery of fertilizer N in the pea-barley intercrop was found to 64%, which was similar to monocropped pea, but slightly less than in monocropped barley (68%). Calculation of RYT-values for the fertilizer N uptake (close to 1), similarly showed that the fertilizer N was used with the same efficiency in the inter- and monocrops.

Effect of intercropping on N_2 fixation in pea

Intercropping pea with barley significantly reduced the amount of N_2 fixed at maturity on a per plant basis, whereas the relative contribution of fixed N_2 to the total N in pea was significantly increased compared to monocropped pea. The increase in the relative contribution of N_2 fixation of the intercropped grain legume have been reported by Danso et al. (1987), Izaurralde et al. (1992) and Tobita et al. (1994). A reduction in the soil inorganic N concentration, due to a fast uptake of inorganic N by the nonlegume, could improve conditions for N_2 fixation, since it is well-known, that there is a negative correlation between inorganic N and amount of N_2 fixed in monocropped pea.

The high competitive ability for soil inorganic N by barley, may be lowered when N fertilizer is added (Vandermeer, 1989), but the shoot/root ratio may be increased and consequently the competition for light between the legume and barley is increased. An increased competition for light may reduce the amount of N_2 fixed per plant compared to monocropped pea, due to a reduced photosynthesis and consequently less photosynthates available for nodule formation and growth and the fixation process in nodules (Wahua and Miller, 1978). However, competition for other environmental sources for plant growth than light, e.g. phosphorus and water, may also cause a reduction in N_2 fixation per plant.

Dinitrogen transfer from pea to barley ?

The ^{15}N enrichment of barley grown in the intercrop was not significantly different from the ^{15}N enrichment of monocropped barley, when compared at the individual harvests. This was due to the large variability in ^{15}N enrichment between replicate plots. Thus, it can be concluded that N transfer could not be detected by the indirect ^{15}N-isotope dilution method in this study. Recent, studies using direct ^{15}N-labeling methods in controlled environments have shown that N transfer between grain legumes and nonlegumes occur, although in small amount (Giller et al., 1991; Hamel et al., 1991; Jensen, 1996a). In an split-root experiment, Jensen (1996a) found that barley obtain up to 19% of its N from intercropped pea, when grown in association for 70 days in a soil with a low inorganic N content. Furthermore, Sawatsky and Soper (1991) and Jensen (1996b) found that significant amounts of N are deposited in the pea rhizosphere during growth.

The inability of the indirect ^{15}N-isotope dilution method to detect N transfer in the field may have several causes: 1) no N-transfer occurred or 2) the resolution of the method is too low. The amount of N transferred from the legume to the nonlegume is often small, causing only a slight dilution of the soil and fertilizer N in the nonlegume plant. Due to the variation often found in ^{15}N enrichment between replicate field plots such small variations are not likely to be significant. Furthermore, the indirect ^{15}N-isotope dilution method only gives an estimate of apparent transfer, since the rooting depth and pattern of N uptake for the nonlegume may be different in the mono- and intercrop, situations. In combination with a declining ^{15}N enrichment of the available soil N pool with time (Witty and Giller, 1991) differences in the ^{15}N enrichments of inter- and monocropped nonlegumes may therefore not be due to a dilution of the soil inorganic N pool by fixed N in the intercrop situation. The results also showed that the ^{15}N content declined whereas the soil N content increased during later growth stages (Figs. 2A and B). The losses of ^{15}N-labeled N may have been due to volatilization of ammonia from the above-ground plant parts (Schjørring et al., 1989). If this process occurs at different rates in the intercrop and monocrop, it will also cause an error when determining N transfer using the indirect ^{15}N-isotope dilution method.

If N transfer occured it could be expected that the amount increased with time, but this was not the case in the experiment in 1984. The results showed that the relative difference in ^{15}N enrichment were highly variable during the late growth stages, which also indicate that differences in ^{15}N-enrichments in the field between intercropped and monocropped barley were too variable to be used for estimating N transfer. Papastylianou and Danso (1991) observed that the estimated N transfer fluctuated between sequential harvests in vetch-oat intercrops. It may be concluded that the indirect ^{15}N-isotope dilution method is not sensitive enough for determining N transfer in annual intercrops in the field.

Nitrogen balance and budget for a pea-barley intercrop

Nitrogen balances were established for the crops supplied with ^{15}N to determine the net effect on the soil N pool of growing monocrops of pea and barley and of the intercrop (Fig. 5). In 3 of the 4 years, the N balance was positive for monocropped pea, whereas it was negative for barley in all years. The intercrop N balance was also negative in all years, but less negative than

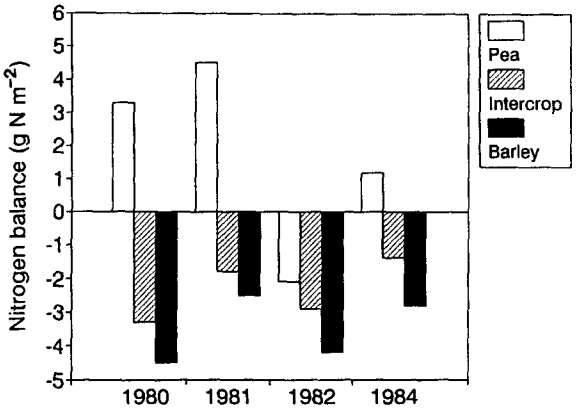


Figure 5. Nitrogen balances for crops supplied with fertilizer N. Balances were calculated according to Equation 8. It is assumed that crop residues are not removed. A negative balance implies that the amount of N removed from the system is higher than the input (N fertilizer and N_2 fixation).

for monocropped barley. The average net balances for N were +1.7 g N m^{-2} for pea, −2.4 for the intercrop and −3.5 g N m^{-2} for the barley monocrop. If fixed N in pea roots was included in the balance it would have been slightly less negative for the intercrop, since any fixed N present in the roots will improve the balance. However, even if fixed N_2 roots are included, intercrops of pea and barley are not likely to increase soil organic N in the long-term, but rather deplete it, as is the case with monocropped barley with low or no N fertilization. The N balance for the unfertilized intercrop is not likely to be less negative, even if the N_2 fixation is larger. The average difference in total N in above-ground plant parts at maturity was found to 4.8 g N m^{-2} of which 90% or more may have been derived from fixation. However, the input of 5 g N m^{-2} as fertilizer N is larger than the increased amount of N fixed and the harvest index for N in intercropped pea was not significantly influenced by fertilization (87%).

Conclusions

The present work showed that intercropping pea and barley without or with a low amount of N fertilizer supply, offers an opportunity to increase the input of symbiotically fixed N_2 into temperate agroecosystem without compromising the yield level and stability. Furthermore, intercropping pea with a cereal seems to be a more optimal cropping strategy in relation to the use of N resources, because the monocropping of pea on fertile soil will result in an overall inefficient use of

N sources. In the monocrop situation the full potential of N_2 fixation in pea may not be exploited, if 50-100 kg N ha^{-1} is available during the growth season. In the intercrop situation the nonlegume will efficiently exploit soil inorganic N sources, while at the same time fixed N_2 enter the plant-soil-system. However, research is needed to develop intercrops with reduced competitive interaction between intercrop components for other environmental sources for plant growth than inorganic N, in order to maximize the input of fixed N and the nonlegume uptake of soil inorganic N sources.

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